

Two-Gate Decision Trust in Inverter-Dominated Power Grids: Gauge-Invariant Model Reducibility, Network–Controller Hybridization, and Intervention Composability

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Abstract—Dynamic-security planning in inverter-dominated grids can fail in two logically independent ways. First, the physical model may be reduced too aggressively: a matrix-valued converter network may not admit independent scalar channels because its frequency-dependent cycle transports are not simultaneously diagonalizable. Second, even if the physical model is adequate, corrective actions may be evaluated too independently: individually beneficial topology and controller changes can close feedback cycles in intervention space and form a jointly harmful portfolio. This paper unifies these two problems into a *two-gate decision-trust framework*. Gate 1 certifies *model reducibility* using a port-reduced matrix return graph, spanning-tree gauge, fundamental-cycle holonomies, a physical-metric non-Abelian defect, and a spectral-gap scalarization bound. Gate 2 certifies *decision reducibility* by lifting finite-rank actions into an intervention Green matrix, from which exact determinant reduction, acyclic composability, strongly-connected-component factorization, an all-order cycle expansion, and a truncation bound follow.

Three bridge results connect the two gates. First, the intervention Green matrix is invariant under consistent local similarity/gauge transformations; hence action feedback cycles are not coordinate artifacts. Second, when the physical resolvent is expanded in network walks, each action-to-action coupling is a coarse-grained sum of matrix-valued physical walks, so closed intervention cycles are “cycles of physical walks.” Third, under exact physical scalarizability, channel-preserving actions, and acyclic normalized action graphs, independent scalar screening is exact at the determinant level; failure of either hypothesis identifies the minimum model/optimization complexity required.

Existing numerical campaigns provide complementary evidence. In a 2,000-pair Laplacian-cospectral study, scalar spectra are identical to numerical precision while 4.35% of full-MIMO pairs disagree on stability. In an IEEE 39-bus low-damped PLL stress case, line 25–38 increases generalized-Fiedler connectivity by 0.1459% and standard Fiedler connectivity by 0.2980%, yet the full coupled mode moves from $-0.01507 + j17.22591$ to $+0.25019 + j19.77892 \text{ s}^{-1}$ while a decoupled model misses the event. Independently, a line reinforcement improves damping by 1.45% and a controller retuning by 5.39%, but their joint application reduces the damping margin by 10.76%. Under co-

retirement of inertia and synchronous damping, pair and triple intervention cycles scale approximately as α^{-1} and $\alpha^{-3/2}$. The resulting engineering message is narrower than a new scalar grid-strength index: *model trust does not imply decision trust*. The two gates determine when scalar screening is defensible, when full MIMO dynamics are needed, when actions may be ranked independently, and when portfolios must be optimized jointly.

Index Terms—inverter-based resources, grid-following converter, grid-forming converter, power-system stability, MIMO networks, graph theory, holonomy, gauge invariance, scalarization, self-energy, nonlinear eigenvalue problem, Green function, low-rank perturbation, intervention portfolio, higher-order interactions.

I. INTRODUCTION

POWER-SYSTEM planning increasingly combines transmission reinforcement, synchronous-machine retirement, grid-forming (GFM) deployment, grid-following (GFL) converter controls, virtual inertia, damping support, and controller retuning. The corresponding planning workflow often uses two simplifying assumptions. The first is a *model assumption*: a detailed multivariable grid can be represented by scalar graph or decoupled impedance channels. The second is a *decision assumption*: candidate upgrades can be evaluated one at a time and then combined according to their individual benefit.

These assumptions answer different questions. A reduced model can be structurally invalid even when every candidate action is weakly coupled. Conversely, a scalar model can be exact while several finite actions interact strongly through the network resolvent. Treating both failures as one generic “converter interaction” problem hides the decision structure that a planner actually needs.

Two recent theory-development tracks motivate a unified view. The first represents a heterogeneous multi-inverter grid as a frequency-dependent matrix graph after port reduction. A spanning-tree gauge trivializes tree transports, leaving fundamental-cycle holonomies as global invariants. Simultaneous diagonalizability of those holonomies determines exact scalarizability, while noncommutativity and Jordan defectiveness provide distinct obstructions. A physical port metric makes the non-Abelian defect coordinate invariant, and a spectral-gap theorem bounds the residual off-diagonal

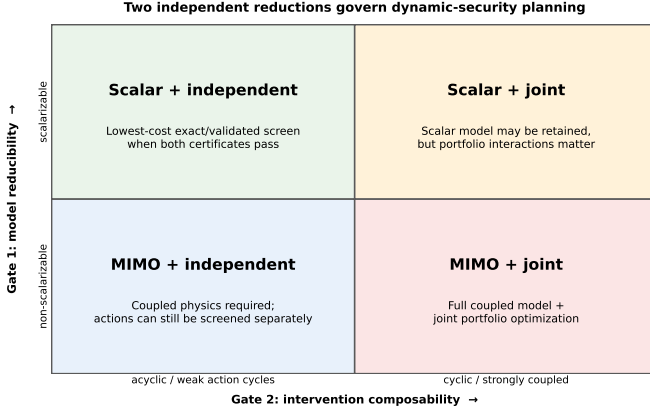


Fig. 1. Two-gate decision-trust map. Model reducibility and intervention composability are logically independent. Passing both gates permits the cheapest representation; failing either gate identifies the extra model or optimization complexity that must be retained.

coupling of the scalarized representation. The same framework yields connectivity–stability counterexamples in which scalar connectivity improves although the full coupled mode degrades.

The second track starts from an exact Schur self-energy bridge that retains condensed controller dynamics in a frequency-dependent operator. It develops network–controller hybridization, controller-aware topology sensitivity, and a finite-dimensional intervention Green matrix. The latter exposes how corrective actions feed back on each other. An acyclic action graph composes exactly; strongly connected components (SCCs) contain all non-composable interaction; and a matrix-log expansion organizes pair, triple, and higher-order action cycles.

The unified question is therefore:

*When is it legitimate to reduce the physical model,
and when is it legitimate to reduce the portfolio
decision?*

This paper proposes a two-gate answer, summarized in Fig. 1. Gate 1 decides whether the physical dynamics can be represented by independent scalar channels. Gate 2 decides whether a set of interventions can be treated independently. The four resulting regimes are operational rather than merely descriptive: scalar/independent, scalar/joint, MIMO/independent, and MIMO/joint.

The main contributions of the synthesis are:

- 1) a common controller-aware matrix operator that supports both physical-cycle and intervention-cycle analysis;
- 2) a gauge-invariance theorem for the action Green matrix and all normalized intervention cycles;
- 3) a network-to-action cycle-lifting interpretation showing that action couplings are coarse-grained sums of physical matrix walks;
- 4) an exact double-reducibility certificate under simultaneous physical scalarization, channel-preserving elementary actions, and acyclic action graphs;
- 5) a calibrated engineering workflow that separates the model-validity gate from the portfolio-composability gate and links each to already observed IEEE 39-bus counterexamples.

The algebraic ingredients—Schur complements, simultaneous diagonalization, determinant lemmas, resolvents, matrix loga-

rithms, and biorthogonal eigenvalue sensitivity—are established mathematics. The candidate contribution is their synthesis into a two-gate planning theory and the decision consequences demonstrated by the existing campaigns. The full-order engineering priority claim remains contingent on replication with documented SG/GFL/GFM or EMT models.

II. COMMON CONTROLLER-AWARE MATRIX MODEL

A. Schur self-energy bridge

After algebraic elimination, partition the linearized dynamic states into retained network/interface coordinates x_r and controller/auxiliary coordinates x_c :

$$\begin{bmatrix} \dot{x}_r \\ \dot{x}_c \end{bmatrix} = \begin{bmatrix} A_{rr} & A_{rc} \\ A_{cr} & A_{cc} \end{bmatrix} \begin{bmatrix} x_r \\ x_c \end{bmatrix}. \quad (1)$$

For $s \notin \text{spec}(A_{cc})$,

$$x_c = (sI - A_{cc})^{-1} A_{cr} x_r, \quad (2)$$

and the exact retained-space operator is

$$S_r(s) = sI - A_{rr} - \Sigma_A(s), \quad (3)$$

$$\Sigma_A(s) = A_{rc}(sI - A_{cc})^{-1} A_{cr}. \quad (4)$$

The block determinant identity gives

$$\det(sI - A) = \det(sI - A_{cc}) \det S_r(s), \quad (5)$$

with the usual analytic continuation at isolated bookkeeping singularities. When retained coordinates admit a second-order electromechanical representation, we write

$$\mathcal{T}(s) = s^2 M + s \Sigma(s) + L. \quad (6)$$

Unlike a fixed nodal damping matrix, $\Sigma(s)$ may be non-diagonal and frequency dependent. For diagonalizable A_{cc} it has a pole-residue form, so each controller family leaves both a spectral and spatial footprint.

B. Port-reduced return graph

An equivalent port representation is

$$F(s)z = 0, \quad F(s) = D(s) - E(s) = D(s)[I - T(s)], \quad (7)$$

with block-diagonal local dynamics $D(s)$ and off-diagonal interconnection blocks $E(s)$. Whenever $D(s)$ is invertible,

$$T(s) = D^{-1}(s)E(s), \quad \det F(s) = \det D(s) \det[I - T(s)]. \quad (8)$$

Thus a collective network mode satisfies $1 \in \text{spec}[T(s)]$. Equation (6) emphasizes controller memory in the retained dynamic stiffness, while (7) exposes the same closed-loop physics as a matrix-valued graph.

III. GATE 1: CAN THE PHYSICAL MODEL BE SCALARIZED?

A. Tree gauge and cycle holonomies

Let each node port space be $V_i \simeq \mathbb{C}^d$ and let a local coordinate change be $z_i = Q_i \tilde{z}_i$. Edge transports transform as

$$T_{ij} \mapsto \tilde{T}_{ij} = Q_i^{-1} T_{ij} Q_j. \quad (9)$$

For an oriented closed walk $C = (i_0, i_1, \dots, i_r = i_0)$ define the cycle holonomy

$$H_C = T_{i_0 i_{r-1}} \cdots T_{i_2 i_1} T_{i_1 i_0}. \quad (10)$$

Then $H_C \mapsto Q_{i_0}^{-1} H_C Q_{i_0}$, so its eigenvalues, trace, determinant, and characteristic polynomial are gauge invariant.

Choose a spanning tree and recursively gauge every invertible tree transport to identity. The remaining chord matrices are the fundamental-cycle holonomies $H_1, \dots, H_\beta$, where $\beta = m - n + 1$.

Gate-1 theorem (exact pointwise scalarizability). A connected equal-port-dimension matrix network with invertible tree transports can be transformed into independent scalar edge channels at a fixed frequency if and only if all fundamental-cycle holonomies are simultaneously diagonalizable. If each H_r is individually diagonalizable over $\mathbb{C}$, this is equivalent to pairwise commutation

$$[H_r, H_q] = 0 \quad \forall r, q. \quad (11)$$

The reason is simple: once every tree edge is I , any further gauge preserving the tree must be the same similarity at every vertex, so one common basis must diagonalize every chord.

For a common scalar basis over a band Ω , the same condition must hold across the frequency-indexed family:

$$[H_r(j\omega), H_q(j\nu)] = 0, \quad \forall r, q, \omega, \nu \in \Omega, \quad (12)$$

plus diagonalizability. A commuting but defective Jordan family is therefore a separate obstruction.

B. Gauge-invariant obstruction and error certificate

Let $G \succ 0$ be a physical port metric, transforming as $G \mapsto Q^* G Q$. Define

$$\|A\|_G^2 = \text{tr}(G^{-1} A^* G A) \quad (13)$$

and the normalized non-Abelian defect

$$\mathcal{N}_{rq} = \frac{\|H_r H_q - H_q H_r\|_G}{\|H_r H_q\|_G + \|H_q H_r\|_G}. \quad (14)$$

This quantity is invariant under arbitrary nonsingular local coordinate changes when the metric is transformed consistently.

To obtain a quantitative near-scalarization statement, whiten the physical metric, choose a reference holonomy $\hat{H}_0$ with simple spectrum, and let

$$S^{-1} \hat{H}_0 S = \Lambda, \quad \delta_0 = \min_{i \neq j} |\lambda_i - \lambda_j|. \quad (15)$$

For another whitened holonomy $\hat{H}_q$, define $B_q = S^{-1} \hat{H}_q S$. Then

$$\|\text{offdiag}(B_q)\|_F \leq \frac{\kappa_2(S)}{\delta_0} \|[\hat{H}_0, \hat{H}_q]\|_F. \quad (16)$$

The bound separates three causes of scalar-model fragility: noncommutativity, small spectral separation, and poor eigenvector conditioning. In 2,000 structured 2×2 matrix networks, the reported campaign found zero bound violations, median bound/actual ratio 1.0659, and 95th percentile 1.3589. Exactly commuting gauge tests scalarized to machine precision.

C. Topology-only nonidentifiability

A stronger falsification uses 2,000 non-isomorphic Laplacian-cospectral graph pairs. Within each pair, the scalar Laplacian spectrum, degree sequence, nodal matrix, and multiset of local 2×2 edge blocks are preserved. The scalarized spectra agree to within 7.27×10^{-14} , yet 4.35% of paired trials produce opposite full-MIMO stability verdicts. Therefore neither the selected scalar topology spectrum nor unchanged local coupling statistics are sufficient statistics for full matrix-network stability.

This result is not an argument that the non-Abelian defect is a universal stability score. Its role is narrower: it certifies when independent scalar channels are structurally unjustified.

IV. GATE 2: CAN CORRECTIVE ACTIONS BE COMPOSED?

A. Finite-rank intervention representation

Let $F_0(s)$ denote the controller-aware base operator and let a portfolio $A = \{a_1, \dots, a_m\}$ be represented as

$$F_A(s) = F_0(s) + U_A(s) C_A(s) V_A^*(s). \quad (17)$$

Line changes, damping/inertia injections, converter retunings, and other localized linearized interventions can be embedded in this form, potentially after decomposing a higher-rank action into elementary low-rank components.

Define the base Green operator and the action-space interaction matrix

$$G_0(s) = F_0^{-1}(s), \quad (18)$$

$$K_A(s) = C_A V_A^* G_0 U_A. \quad (19)$$

The matrix determinant lemma yields the exact reduction

$$\frac{\det F_A(s)}{\det F_0(s)} = \det[I + K_A(s)]. \quad (20)$$

Thus full-grid physics remains in G_0 , while an m -action portfolio is represented by an $m \times m$ nonlinear interaction matrix.

B. Exact composability, SCCs, and closed cycles

Write

$$K = D_K + E_K, \quad D_K = \text{diag}(K), \quad E_K = \text{offdiag}(K), \quad (21)$$

and normalize interactions by

$$Q = (I + D_K)^{-1} E_K. \quad (22)$$

Then

$$\det(I + K) = \left[\prod_{a \in A} (1 + K_{aa}) \right] \det(I + Q). \quad (23)$$

The first factor contains isolated-action effects; $\det(I + Q)$ is the pure portfolio correction.

Gate-2 theorem (acyclic composability). If the directed graph induced by Q is acyclic, a topological permutation makes Q strictly upper triangular. Hence

$$\det(I + Q) = 1, \quad (24)$$

and the portfolio determinant is exactly the product of its isolated-action factors.

If the action graph contains cycles, permuting Q into SCC condensation form gives

$$\det(I + Q) = \prod_{c \in \text{SCC}} \det(I + Q_c). \quad (25)$$

All non-composable determinant effects are therefore confined to feedback cores, which need not correspond to physically adjacent assets.

If $\rho(Q) < 1$,

$$\log \det(I + Q) = \sum_{p=1}^{\infty} \frac{(-1)^{p+1}}{p} \text{tr}(Q^p). \quad (26)$$

With zero diagonal, the first nonzero term is the pair-cycle contribution

$$-\frac{1}{2} \text{tr}(Q^2) = -\sum_{i < j} q_{ij} q_{ji}, \quad (27)$$

followed by genuine three-action cycles from $\text{tr}(Q^3)$. For $q = \|Q\|_2 < 1$, truncating after order p gives

$$|R_p| \leq \frac{mq^{p+1}}{(p+1)(1-q)}. \quad (28)$$

This provides an interaction-order gate: individual actions suffice only when the certified residual is small enough for the planning tolerance.

V. BRIDGE I: GAUGE-INVARIANT INTERVENTION CYCLES

The physical-cycle theory requires coordinate-invariant statements. The intervention graph should obey the same standard; otherwise a “strong action cycle” could be an artifact of state units or local controller coordinates.

Theorem 1 (gauge invariance of the action Green matrix). Let a nonsingular similarity $z = S\tilde{z}$ transform the base operator and an action portfolio consistently as

$$F'_0 = S^{-1} F_0 S, \quad (29)$$

$$U'_A = S^{-1} U_A, \quad (30)$$

$$V'_A = V_A^* S, \quad (31)$$

with the action amplitude matrix C_A unchanged. Then

$$K'_A = K_A. \quad (32)$$

Consequently Q , its directed action graph, its SCC decomposition, $\det(I + Q)$, $\text{tr}(Q^p)$, and every closed action-cycle product are invariant.

Proof. Since $G'_0 = F'^{-1}_0 = S^{-1} G_0 S$,

$$K'_A = C_A V'^*_A G'_0 U'_A \quad (33)$$

$$= C_A V'^*_A S (S^{-1} G_0 S) S^{-1} U_A \quad (34)$$

$$= C_A V'^*_A G_0 U_A = K_A. \quad (35)$$

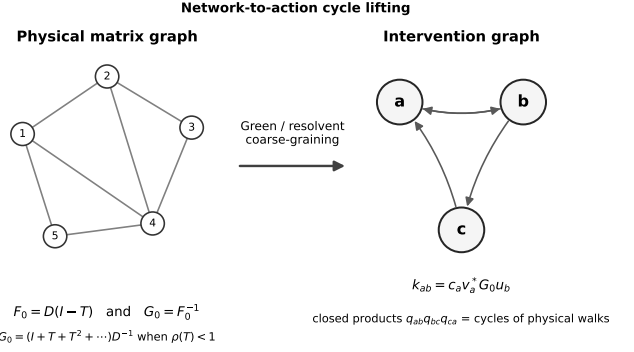


Fig. 2. Network-to-action cycle lifting. The physical resolvent sums matrix-valued network walks. Each action-to-action element k_{ab} aggregates the walks that transmit intervention b into intervention a ; closed products in Q are therefore coarse-grained cycles of physical walks.

All listed quantities are functions of the unchanged matrix K_A . $\square$

This bridge is useful because it places physical holonomy cycles and action cycles on the same invariance footing: both describe closed feedback structure that survives admissible changes of local representation.

VI. BRIDGE II: NETWORK-TO-ACTION CYCLE LIFTING

The Green operator in (19) provides a direct algebraic bridge from matrix-valued network walks to intervention feedback.

From (7),

$$G_0(s) = F_0^{-1}(s) = [I - T(s)]^{-1} D^{-1}(s). \quad (36)$$

Whenever $\rho(T) < 1$ (or on any domain where the corresponding resolvent series is valid),

$$G_0 = (I + T + T^2 + \dots) D^{-1}. \quad (37)$$

The (i, j) block of T^ℓ is the sum of ordered matrix products over all length- ℓ walks from port j to port i . For elementary rank-one actions a and b ,

$$k_{ab} = c_a v_a^* G_0 u_b = \sum_{\ell=0}^{\infty} c_a v_a^* T^\ell D^{-1} u_b. \quad (38)$$

Therefore k_{ab} is not merely a statistical “interaction score.” It is a coarse-grained sum of the physical matrix walks through which action b propagates into the response direction associated with action a .

A closed action product such as

$$k_{ab} k_{bc} k_{ca} \quad (39)$$

concatenates three such Green-mediated walk families. Expanding each factor with (38) produces a weighted sum of closed alternating walks between *action space* and *physical network space*. This is illustrated in Fig. 2.

This interpretation also explains why action-space proximity is not the same as geographic proximity. Two actions can be electrically remote yet strongly coupled if the relevant modal Green function has large biorthogonal transfer between their support directions.

VII. BRIDGE III: A DOUBLE-REDUCIBILITY CERTIFICATE

The two gates can now be combined into an exact sufficient certificate for the cheapest planning representation.

Let Gate 1 produce a common scalar-channel basis at a fixed s . Suppose every physical intervention is decomposed into elementary action components that preserve those scalar channels. Let $K^{(r)}$ be the action interaction matrix inside channel r , with normalized off-diagonal matrix $Q^{(r)}$.

Theorem 2 (double reducibility). Assume:

- 1) the physical matrix graph is exactly scalarizable in one common basis at the frequency of interest;
- 2) every elementary intervention is channel preserving in that basis, so the transformed portfolio operator is a direct sum over scalar channels;
- 3) for every channel r , the directed graph induced by $Q^{(r)}$ is acyclic.

Then the normalized portfolio determinant factorizes exactly into individual scalar-channel action factors:

$$\frac{\det F_A}{\det F_0} = \prod_r \prod_{a \in A_r} (1 + K_{aa}^{(r)}). \quad (40)$$

Hence independent scalar screening is exact at the determinant level under these hypotheses.

Proof. Gate 1 and channel-preserving actions make the transformed full portfolio operator block diagonal across scalar channels, so its determinant is the product of channel determinants. Within each channel, the finite-rank determinant reduction applies. Acyclicity implies $\det(I + Q^{(r)}) = 1$, leaving only diagonal individual-action factors. Multiplying over channels gives (40). $\square$

The contrapositive is not a universal instability statement. Rather, failure of the hypotheses identifies what may not be discarded:

- Gate 1 fails: retain MIMO/coupled physical dynamics;
- Gate 2 fails: retain joint portfolio interactions;
- both fail: use a coupled physical model and optimize the action SCCs jointly.

This is the formal basis for the four regimes in Fig. 1.

VIII. NETWORK-CONTROLLER HYBRIDIZATION AND TOPOLOGY SENSITIVITY

The self-energy representation explains why Gate 1 can become most fragile near controller/network modal coalescence and why local topology changes can have nonlocal signs.

A. Scale-invariant mode-family participation

A naive controller-state norm fraction changes under state-unit rescaling. For right and left eigenvectors r, l , define instead

$$\pi_c^{\text{bio}} = \frac{\sum_{i \in G} |\bar{l}_i r_i|}{\sum_i |\bar{l}_i r_i|}. \quad (41)$$

Under an invertible diagonal state scaling $x' = Sx$, right and left eigenvectors transform as $r' = Sr$ and $l' = S^{-H}l$, so each product $\bar{l}_i r_i$ is invariant. In the reported coordinate-scaling test, changing controller-state scale from 10^{-6} to 10^6 drove the norm-based fraction from near zero to near one while the biorthogonal fraction remained 0.597033.

B. Two-mode hybridization

For a local network/controller pair

$$H_{nc} = \begin{bmatrix} \lambda_n & g_{nc} \\ g_{cn} & \lambda_c \end{bmatrix}, \quad (42)$$

the coupled poles are

$$\lambda_{\pm} = \frac{\lambda_n + \lambda_c}{2} \pm \sqrt{\frac{(\lambda_n - \lambda_c)^2}{4} + g_{nc}g_{cn}}. \quad (43)$$

A useful reduced hybridization indicator is

$$\mathcal{H}_{nc} = \frac{4|g_{nc}g_{cn}|}{|\lambda_n - \lambda_c|^2}. \quad (44)$$

This is not a universal stability margin; it diagnoses loss of a clean network/control family separation. Near the reported coalescence point, the two-mode pole reduction had sub-1% maximum error over the sweep, modal condition number reached approximately 73.5, and the maximum topology sensitivity was about 50 times the detuned value.

C. Controller-aware topology derivative

For a simple nonlinear eigenvalue satisfying

$$\mathcal{T}(s, \rho)v = 0, \quad y^* \mathcal{T}(s, \rho) = 0, \quad (45)$$

the exact sensitivity is

$$\frac{ds}{d\rho} = -\frac{y^* \mathcal{T}_\rho v}{y^* \mathcal{T}_s v}. \quad (46)$$

For (6),

$$\frac{ds}{d\rho} = -\frac{y^*(s^2 M_\rho + s \Sigma_\rho + L_\rho)v}{y^*(2sM + \Sigma + s \Sigma_s)v}. \quad (47)$$

If a line parameter also moves the solved AC operating point x^* through $F_{\text{pf}}(x^*, z, P, Q) = 0$, implicit differentiation separates

$$\frac{d\mathcal{T}}{dz_e} = \underbrace{L_{z_e}}_{\text{structural}} - \underbrace{L_x F_x^{-1} F_{z_e}}_{\text{flow-mediated}} + s \underbrace{(\Sigma_{z_e} - \Sigma_x F_x^{-1} F_{z_e})}_{\text{controller-mediated}}. \quad (48)$$

This decomposition supports controller-resonant topology reversals. In one mechanism case, line 25–26 changes from a network-only damping derivative $+1.3408 \times 10^{-4}$ to a controller-aware derivative -4.2715×10^{-3} , a sign reversal and magnitude increase of about 31.9.

IX. GATE-1 EVIDENCE: IEEE 39-BUS CONNECTIVITY-STABILITY REVERSAL

The AC-consistent reduced IEEE 39-bus campaign provides a falsification-oriented Gate-1 example. For every candidate line, nonlinear AC power flow was re-solved, the P, Q injection Jacobian was recomputed, load buses were eliminated, and SG/GFL/GFM reduced dynamics were connected to the ten retained generator ports. Four low-inertia GFL placements were located at buses 30, 34, 37, and 38.

Under nominal PLL tuning ($K_p = 25$, $K_i = 400$), none of 180 candidate line additions caused instability. This negative result is important: ordinary reinforcement is not claimed to be generically destabilizing.

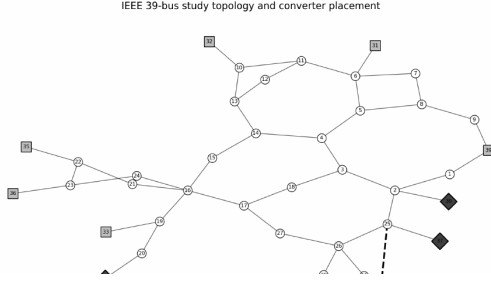


Fig. 3. IEEE 39-bus reduced study topology from the scalarizability campaign. Generator buses are dynamic ports after load-bus reduction; buses 30, 34, 37, and 38 are the low-inertia GFL placements, and line 25–38 is the flagship Gate-1 intervention.

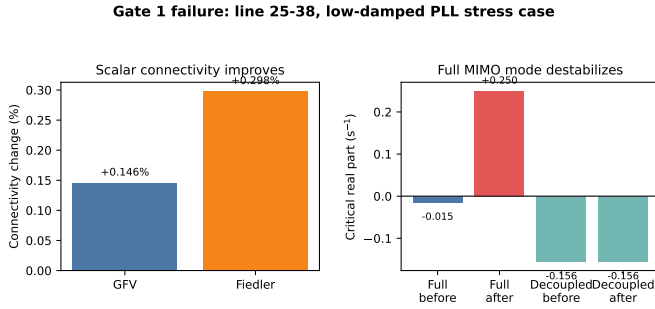


Fig. 4. Gate-1 counterexample. Scalar connectivity improves under line 25–38 while the full coupled mode becomes unstable; the decoupled model remains on the stable side and misses the relevant mode family.

Under a deliberately low-damped PLL stress case ($K_p = 1$, $K_i = 400$), the baseline remains stable at

$$s_0 = -0.01507 + j17.22591 \text{ s}^{-1}. \quad (49)$$

Adding line 25–38 gives

$$\mu_2^{\text{GFV}} : 0.2349169 \rightarrow 0.2352597 \quad (+0.1459\%), \quad (50)$$

$$\lambda_2(K_r) : 12.08886 \rightarrow 12.12489 \quad (+0.2980\%), \quad (51)$$

while the full coupled critical mode becomes

$$s_{\text{crit}} = +0.25019 + j19.77892 \text{ s}^{-1}. \quad (52)$$

The crossing occurs at line-strength factor $\alpha_c = 0.21298$. A deliberately decoupled approximation predicts essentially no adverse change, from $-0.15627 + j7.30340$ to $-0.15625 + j7.30515 \text{ s}^{-1}$.

Cycle diagnostics support the modeling interpretation rather than being used as stability proxies. The compatible cycle-scalarization residual increases from 0.0673 before the upgrade to 0.1733 at the stability crossing. At the crossing, the full return graph satisfies unit return to approximately 5.4×10^{-15} while the cycle-scalarized representation remains 0.296 away. The exact post-upgrade unit-return decomposition is collective: the largest fundamental closure carries only 8.84% of total chord magnitude and the effective number of participating closures is 28.9.

Biorthogonal participation localizes the unstable mode mainly at GFL buses 30, 37, 38, and 34. Retuning only the

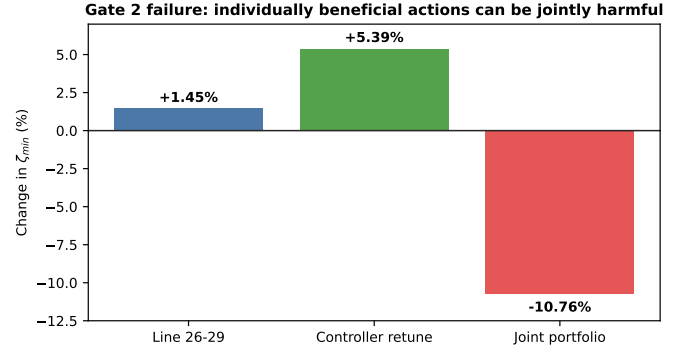


Fig. 5. Gate-2 counterexample. Both actions are damping-positive when screened alone, yet the joint portfolio is damping-negative relative to baseline.

proportional PLL gain at bus 30 yields a zero crossing at $K_{p,30}^* = 2.10837$. With $K_{p,30} = 3$ the same upgraded grid has

$$s_{\text{crit}} = -0.10064 + j19.62628 \text{ s}^{-1}, \quad (53)$$

while the connectivity improvement remains unchanged. The reversal is controller specific: nominal GFL tuning and the tested GFM surrogates remain stable in the corresponding upgrade tests.

X. GATE-2 EVIDENCE: DESTRUCTIVE CORRECTIVE-ACTION PORTFOLIOS

A separate IEEE 39-bus mechanism campaign isolates the decision-reducibility problem. Across 17,664 line/controller action pairs, a determinant-based spectral interaction observable had Spearman correlation 0.673 with damping non-additivity and AUC 0.897 for identifying the strongest 5% interactions. A separate 23,000 line–GFM-like pair campaign reported Spearman 0.747 and AUC 0.894. The hybridization indicator itself had Spearman 0.920 with the maximum mixed-action interaction across operating points.

The flagship portfolio has baseline damping margin

$$\zeta_{\min}^{(0)} = 0.045218. \quad (54)$$

Reinforcing line 26–29 alone gives +1.45%, and retuning the controller natural frequency from 5.1 to 4.8 rad/s alone gives +5.39%. The joint action instead gives

$$\Delta\zeta_{\min}^{\text{joint}} = -10.76\%, \quad (55)$$

with mixed non-additivity $\mu_{ec} = -7.96 \times 10^{-3}$. The hybridization indicator rises to 1.121 jointly. Figure 5 shows the decision consequence.

Topology-only portfolios show the same structural phenomenon. In one high-IBR case, line 19–20 alone improves $\zeta_{\min}$ by 15.8% and line 20–34 alone by 9.1%, yet their joint reinforcement reduces the margin by 7.55% below baseline. The three highest-ranked individual line actions together reduce the margin by 6.87%, whereas an interaction-aware triple improves it by 19.59%. A pair Green kernel had Spearman 0.772 with the magnitude of pair Moebius interaction and AUC about 0.96 for detecting the strongest 5%; a three-cycle kernel had Spearman 0.874 with the genuine third-order interaction.

These observations do not imply that every high-IBR portfolio is non-composable. Rather, they demonstrate that

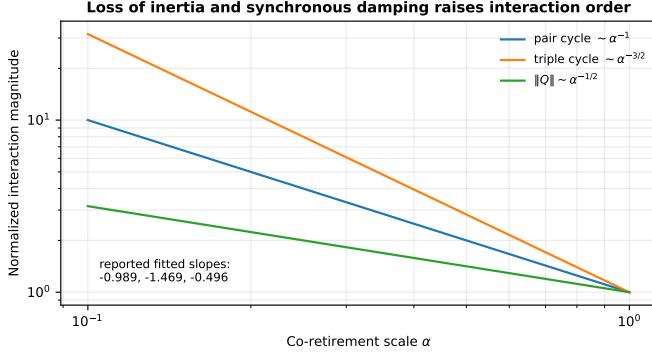


Fig. 6. Reported low-inertia/damping scaling law under uniform co-retirement. The plotted curves show the theoretical exponents; annotations give the corresponding fitted slopes from the mechanism campaign.

independent ranking has no universal guarantee for the tested damping-margin set function, which exhibits both positive and negative interactions and is not universally monotone, submodular, or supermodular over the tested regime.

XI. LOW-INERTIA/DAMPING SCALING OF DECISION COMPLEXITY

A high-risk hypothesis was that lower inertia alone should always make action portfolios more strongly coupled. The mechanism campaign falsified that universal statement: the trend depends jointly on inertia and damping.

For an isolated dominant mode with modal mass m_k , damping d_k , and stiffness λ_k , consider uniform co-retirement

$$M_\alpha = \alpha M_0, \quad D_\alpha = \alpha D_0. \quad (56)$$

The natural frequency scales as $\omega_k \propto \alpha^{-1/2}$. At $s = j\omega_k$, inertia and stiffness cancel in modal dynamic stiffness and

$$|G_k(j\omega_k)| \propto \alpha^{-1/2}. \quad (57)$$

A closed action cycle of order p with p Green factors therefore obeys

$$|K^{(p)}| \propto \alpha^{-p/2}. \quad (58)$$

Thus pair and triple cycles scale as α^{-1} and $\alpha^{-3/2}$. The reported fitted slopes were -0.989 and -1.469 , while $\|Q\|$ scaled with slope -0.496 , all with $R^2 > 0.99989$ in the isolated-mode campaign.

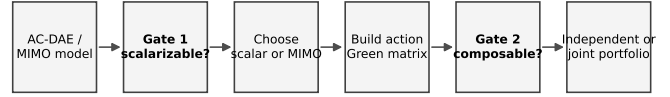
If inertia decreases while absolute damping is preserved, the Green magnitude instead scales as $\alpha^{+1/2}$ and the cycle exponents reverse sign. The engineering implication is not “low inertia is always bad.” It is that simultaneous loss of inertial and dissipative support can increase the order of portfolio interaction that must be retained.

XII. UNIFIED PLANNING WORKFLOW

Figure 7 summarizes the proposed workflow.

Step 1: solve and retain the correct physics. Solve the AC operating point and obtain a local matrix port model $F(s)$ analytically, through DAE elimination, or from frequency-response data. Preserve controller self-energy rather than collapsing it automatically into fixed nodal damping.

Two-gate dynamic-security planning workflow



Validate final portfolios in full nonlinear / EMT models, including limits, protections, voltage and N-1 constraints.

Fig. 7. Two-gate workflow. Gate 1 determines the minimum physical-model complexity; Gate 2 determines whether candidate interventions may be screened independently or must be optimized in coupled SCC portfolios.

Step 2: apply Gate 1. Construct $T(s)$, a spanning-tree gauge, and the fundamental-cycle holonomies over the relevant frequency band. Carry a physical port metric. Test commutators, diagonalizability/Jordan defect, eigenvector conditioning, and the error certificate (16). If the required scalarization certificate fails, retain the smallest MIMO representation that preserves the coupled cycle family.

Step 3: inspect hybridization and topology sensitivity. Identify the margin-setting mode using biorthogonal participation. Near network/controller coalescence, use the nonlinear-eigenvalue derivative (47) and, when necessary, separate structural, flow-mediated, and controller-mediated topology channels.

Step 4: apply Gate 2. Encode candidate line, inertia, damping, BESS/GFM, and controller actions in $U_A C_A V_A^*$. Form K_A and Q . If the action graph is acyclic, isolated determinant effects compose exactly. Otherwise, decompose into SCCs and use (28) to choose the minimum certified interaction order.

Step 5: optimize only the necessary coupled cores. Weak or acyclic action sets may be ranked independently. Strongly connected action cores are optimized jointly. This avoids an all-to-all combinatorial study while preventing the false assumption that all actions are independent.

Step 6: full validation. Validate only final candidate portfolios using the complete small-signal and nonlinear/EMT models with current limits, protections, voltage/reactive constraints, and $N - 1$ feasibility. The two gates are screening and structure tools, not substitutes for final certification.

XIII. DECISION-TRUST TAXONOMY AND INDUSTRIAL INTERPRETATION

Table I translates the theory into a planning taxonomy.

The immediate industrial value is not another universal “grid strength” number. It is a complexity certificate. Examples include:

- coordinating transmission reinforcement with GFM/BESS installation;
- retuning PLL, droop, VSM, or damping loops after topology changes;
- assessing synchronous-machine retirement where inertia and damping disappear together;
- identifying which candidate portfolios deserve expensive EMT studies;

TABLE I
TWO-GATE DECISION-TRUST TAXONOMY

	Gate 2 passes: acyclic/weak portfolio	Gate 2 fails: cyclic/strong portfolio	Planning interpretation
Gate 1 passes: scalarizable physics	Independent scalar screening	Scalar physical channels, but joint action optimization	Physical reduction is defensible; the remaining question is whether actions feed back on each other.
Gate 1 fails: non-scalarizable physics	Coupled MIMO model with independent action screening	Coupled MIMO model plus SCC/joint portfolio optimization	Scalar graph/decoupled analysis is structurally incomplete; action coupling may or may not add a second layer of complexity.

- distinguishing a bad asset from a bad *combination* of otherwise useful assets.

The flagship Gate-1 example shows why “more connected” is not equivalent to “more stable” in a coupled converter network. The Gate-2 example shows why “two good actions” is not equivalent to “one good portfolio.” Taken together, these produce the central engineering statement:

$$\boxed{\text{Model trust does not imply decision trust.}} \quad (59)$$

XIV. CLAIM CALIBRATION, FALSIFICATION TESTS, AND LIMITATIONS

Table II separates established mathematical infrastructure from candidate contributions and evidence status.

Several limitations are deliberate. First, the strongest IEEE 39-bus results use reduced or mechanism-level converter models rather than manufacturer-grade control stacks. Second, $\rho(Q) > 1$ marks loss of the standard cycle-series convergence condition; it is not itself a stability boundary. Third, the isolated-mode scaling law (58) assumes modal dominance and fixed action normalization. Fourth, Gate 1 diagnoses model reducibility, not stability; a non-Abelian network can be stable. Fifth, Gate 2 diagnoses determinant non-composability, not economic optimality; engineering constraints can alter the final portfolio choice.

The decisive full-order falsification campaign should test, on one common documented SG/GFL/GFM or EMT benchmark:

- 1) whether a connectivity-improving intervention can produce a coupled mode missed by a scalar/decoupled reduction;
- 2) whether the Gate-1 scalarization certificate correctly predicts where that reduced model loses trust;
- 3) whether individually beneficial topology/controller/BESS actions can become jointly detrimental;
- 4) whether the action Green matrix and its low-order cycles predict the strongest finite-action non-additivities;
- 5) whether the four two-gate quadrants correctly predict the minimum model and portfolio complexity needed for an engineering decision.

Failure of these tests should narrow the industrial claims without invalidating the exact algebraic identities.

XV. CONCLUSION

This paper unifies two reductions that are routinely conflated in inverter-dominated grid planning. The first asks whether the *physics* can be reduced from a matrix-valued network to independent scalar channels. The second asks whether

the *decision* can be reduced from a coupled action portfolio to isolated rankings. Fundamental-cycle holonomies, gauge invariance, and scalarization bounds answer the first question; the intervention Green matrix, SCC factorization, and closed action cycles answer the second.

The new bridge results show that these are not disconnected theories. The action Green matrix is invariant under consistent local gauge changes. Its entries are coarse-grained sums of physical matrix walks, so intervention cycles are cycles of physical walks. When the physical graph is exactly scalarizable, actions preserve those channels, and each normalized channel action graph is acyclic, independent scalar screening is exact at the determinant level. Otherwise, the failed gate identifies the missing complexity.

The existing IEEE 39-bus evidence illustrates both failure modes. A connectivity-improving line can destabilize a full coupled GFL mode while a decoupled model misses the event. Separately, a line and controller retuning that improve damping individually can reduce the margin by 10.76% when implemented together. Under simultaneous loss of inertia and damping, the importance of higher-order action cycles increases according to a simple scaling law.

The resulting claim is intentionally practical and falsifiable: a dynamic-security planner should certify *model reducibility* before trusting scalar screens and certify *decision reducibility* before adding individual upgrade benefits. The purpose of the two-gate framework is not to replace detailed stability studies, but to determine which detailed studies are actually necessary and where independent planning logic is unsafe.

APPENDIX A

PROOF DETAIL FOR THE PHYSICAL-METRIC INVARIANCE

For $A' = Q^{-1}AQ$ and $G' = Q^*GQ$,

$$\|A'\|_{G'}^2 = \text{tr}(G'^{-1}A'^*G'A') \quad (60)$$

$$= \text{tr}(Q^{-1}G^{-1}(Q^*)^{-1}Q^*A^*(Q^{-1})^*Q^*GQQ^{-1}AQ) \quad (61)$$

$$= \text{tr}(Q^{-1}G^{-1}A^*GAQ) \quad (62)$$

$$= \text{tr}(G^{-1}A^*GA) = \|A\|_G^2. \quad (63)$$

The same identity applies to $[H_r, H_q]$ and each denominator term of (14).

TABLE II
CLAIM CALIBRATION AND EVIDENCE STATUS

Item	Status	Evidence in current synthesis	Claim level
Schur complements, resolvents, NEP sensitivity, determinant lemma, matrix-log expansion	Established mathematics	Exact identities and numerical implementation checks in the source campaigns	Not claimed as new mathematics
Physical cycle scalarizability by simultaneous diagonalization	Standard linear algebra specialized to matrix power networks	2,000 structured cases; commuting cases exact to machine precision	Candidate power-network structural formulation
Physical-metric non-Abelian defect and spectral-gap scalarization bound	Invariance and perturbation bound derived in source scalarizability work	Zero violations in 2,000 structured cases; median slack 6.6%	Candidate methodological/theoretical contribution
Topology-only nonidentifiability	Constructive use of cospectral graph pairs	2,000 pairs, 4.35% opposite full-MIMO stability verdicts	Strong counterexample class
Action Green matrix, acyclic composability, SCC factorization	Finite-rank perturbation theory plus graph factorization	Exact determinant identities; random numerical verification in source campaign	Candidate power-systems portfolio theory
All-order action cycles and truncation bound	Matrix-log infrastructure established; action interpretation is specialized	4,800 truncation tests, no reported bound violations	Candidate theoretical contribution
Gauge invariance of K_A and action cycles	Algebraic bridge developed in this synthesis	Exact proof in Sec. V	Candidate bridge theorem; requires literature priority audit
Network-to-action cycle lifting	Resolvent-walk expansion established; intervention interpretation new to this synthesis	Exact expansion where resolvent series converges	Candidate structural synthesis
Double-reducibility certificate	Consequence of Gate-1 scalarization plus Gate-2 acyclicity	Exact proof under stated channel-preserving assumptions	Candidate two-gate theorem
IEEE 39 Gate-1 reversal	AC-consistent reduced SG/GFL/GFM surrogate	GFV/Fiedler improve; full coupled mode unstable; decoupled model misses event	Strong proof-of-concept, not vendor-grade validation
IEEE 39 Gate-2 destructive portfolio	Mechanism-level controller model	+1.45%, +5.39%, jointly −10.76%	Strong mechanism-level planning counterexample

APPENDIX B

PROOF DETAIL FOR THE SCALARIZATION BOUND

In the reference eigenbasis, $[\Lambda, B_q]_{ij} = (\lambda_i - \lambda_j)(B_q)_{ij}$. Therefore

$$\|[\Lambda, B_q]\|_F^2 \geq \delta_0^2 \|\text{offdiag}(B_q)\|_F^2. \quad (64)$$

Since $[\Lambda, B_q] = S^{-1}[\hat{H}_0, \hat{H}_q]S$,

$$\|[\Lambda, B_q]\|_F \leq \kappa_2(S) \|\hat{H}_0, \hat{H}_q\|_F, \quad (65)$$

which yields (16).

APPENDIX C

PAIR AND TRIPLE ACTION-CYCLE STRUCTURE

For two elementary actions,

$$K = \begin{bmatrix} k_{aa} & k_{ab} \\ k_{ba} & k_{bb} \end{bmatrix}, \quad (66)$$

and

$$\det(I + K) = (1 + k_{aa})(1 + k_{bb}) - k_{ab}k_{ba}. \quad (67)$$

The exact pair correction is therefore the closed feedback product $k_{ab}k_{ba}$. For three actions the determinant additionally contains oriented products $k_{ab}k_{bc}k_{ca}$ and $k_{ac}k_{cb}k_{ba}$, matching the $\text{tr}(Q^3)$ term of the all-order expansion.

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